



HAAT — Entity Relationship Diagram (ERD)

Bangladeshi Multi-Vendor Artisanal E-Commerce Platform • MySQL Schema Specification

Source File: **db.sql** | Target DB: **haat_db**

Entities: **13 Tables** | Engine: **InnoDB (UTF8MB4)** | Version: **2.0 Standard ANSI**



Identity & Workshop
(users, sellers)



Catalog & Taxonomy (categories,
brands, products, reviews)



Orders & Logistics (orders,
order_items, tracking_events)



Social & Marketing (wishlists, coupons,
messages, notifications)

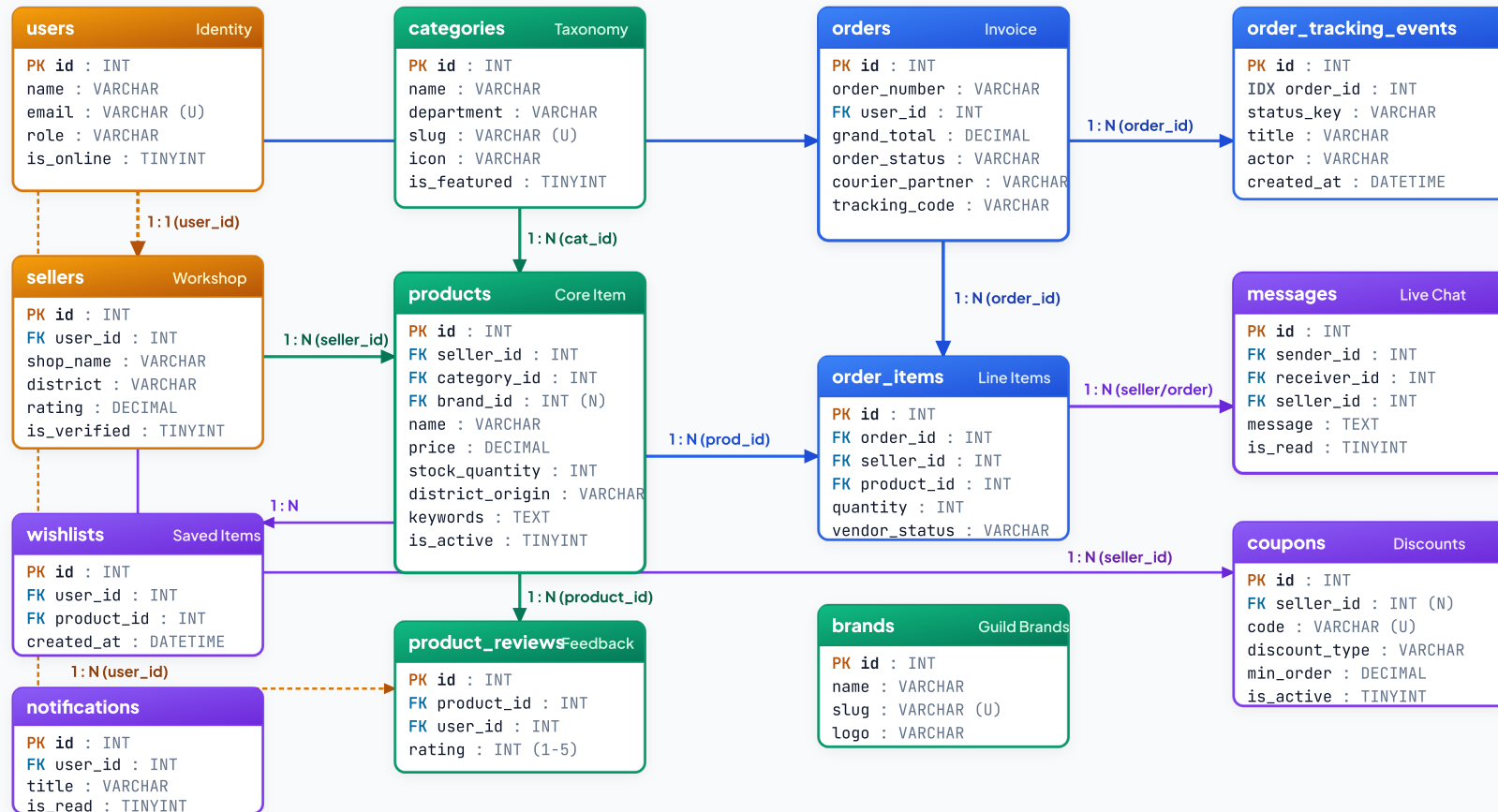


Primary
Key



Foreign
Key

—(1:N)—
Crows Foot
Relation



1. users (Platform Accounts)

IDENTITY

FIELD	TYPE	KEY	NULL	DEFAULT	DESCRIPTION
id	INT	PK	No	AUTO	Primary Key identifier
name	VARCHAR(100)	-	No	-	Full name of user
email	VARCHAR(120)	UNI	No	-	Login email (Unique)
password	VARCHAR(255)	-	No	-	BCrypt hashed password
role	VARCHAR(30)	-	Yes	'customer'	admin / seller / customer
status	VARCHAR(30)	-	Yes	'active'	active / suspended / pending
is_online	TINYINT(1)	-	No	0	Real-time active presence
created_at	DATETIME	-	Yes	CURRENT	Registration timestamp

2. sellers (Artisan Workshops)

WORKSHOP

FIELD	TYPE	KEY	NULL	DEFAULT	DESCRIPTION
id	INT	PK	No	AUTO	Primary Key identifier
user_id	INT	FK	No	-	users.id (ON DELETE CASCADE)
shop_name	VARCHAR(150)	-	No	-	Artisan workshop title
shop_slug	VARCHAR(160)	UNI	No	-	Unique URL storefront slug
district	VARCHAR(50)	-	Yes	'Dhaka'	Craft district (e.g. Tangail)
commission_rate	DECIMAL(5,2)	-	Yes	5.00	Platform commission %
is_verified	TINYINT(1)	-	Yes	1	Admin verified badge
rating	DECIMAL(3,2)	-	Yes	4.80	Aggregate rating score

3. categories (Departmental Hierarchy)

TAXONOMY

FIELD	TYPE	KEY	NULL	DEFAULT	DESCRIPTION
id	INT	PK	No	AUTO	Primary Key identifier
name	VARCHAR(100)	-	No	-	Category name (English)
name_bn	VARCHAR(100)	-	Yes	NULL	Bangla localized label
department	VARCHAR(50)	-	Yes	'Clothes'	Clothes / Food / Art & Accessories
slug	VARCHAR(110)	UNI	No	-	URL SEO slug
icon	VARCHAR(50)	-	Yes	'bi-tag'	Bootstrap icon class
is_featured	TINYINT(1)	-	Yes	1	Featured in homepage

4. products (Artisanal Inventory)

CORE ITEM

FIELD	TYPE	KEY	NULL	DEFAULT	DESCRIPTION
id	INT	PK	No	AUTO	Primary Key identifier
seller_id	INT	FK	No	-	sellers.id (ON DELETE CASCADE)
category_id	INT	FK	No	-	categories.id (ON DELETE CASCADE)
brand_id	INT	FK	Yes	NULL	brands.id (ON DELETE SET NULL)
name	VARCHAR(200)	-	No	-	Product title
price	DECIMAL(10,2)	-	No	-	Regular selling price
sale_price	DECIMAL(10,2)	-	Yes	NULL	Discounted price
stock_quantity	INT	-	No	10	Available warehouse stock
district_origin	VARCHAR(50)	-	Yes	'Dhaka'	District heritage tag
keywords	TEXT	-	Yes	NULL	SEO & search match terms

5. orders & order_items

COMMERCE

FIELD	TYPE	KEY	DESCRIPTION
orders.id	INT	PK	Order unique ID
order_number	VARCHAR(30)	UNI	e.g. HT-2026-9041
user_id	INT	FK	users.id
grand_total	DECIMAL(10,2)	-	Final billed amount
courier_partner	VARCHAR(100)	-	e.g. Pathao Courier
tracking_code	VARCHAR(100)	-	Consignment code
order_items.id	INT	PK	Line item ID
order_id	INT	FK	orders.id
seller_id	INT	FK	sellers.id
product_id	INT	FK	products.id

6. tracking & messages

LIVE SUITE

FIELD	TYPE	KEY	DESCRIPTION
order_tracking_events.id	INT	PK	Timeline log ID
order_id	INT	IDX	Order reference
status_key	VARCHAR(50)	-	pending, craft shipped
actor	VARCHAR(150)	-	System / Artist / Courier
messages.id	INT	PK	Chat message ID
sender_id	INT	FK	users.id (Sender)
receiver_id	INT	FK	users.id (Receiver)
seller_id	INT	FK	sellers.id (Workshop)
message	TEXT	-	Message body text
is_read	TINYINT(1)	-	Read / acknowledged

7. coupons, wishlists, reviews

ENGAGEMENT

FIELD	TYPE	KEY	DESCRIPTION
coupons.id	INT	PK	Discount voucher ID
seller_id	INT	FK	sellers.id (or NULL global)
code	VARCHAR(30)	UNI	Voucher promo code
wishlists.id	INT	PK	Saved bookmark ID
user_id, product_id	INT	UNI	Unique bookmark pair
product_reviews.id	INT	PK	Feedback review ID
rating	INT	-	1 to 5 star rating
notifications.id	INT	PK	System notification ID

Complete Foreign Key & Cardinality Relationship Matrix						REFERENTIAL INTEGRITY
PARENT TABLE	PARENT PK	CHILD TABLE	FOREIGN KEY COLUMN	CARDINALITY	ON DELETE ACTION	BUSINESS RULE
users	id	sellers	user_id	1:1	CASCADE	A vendor account belongs to exactly one user identity. Deleting user removes shop.
users	id	orders	user_id	1:N	CASCADE	A registered customer can place multiple historical purchase orders.
sellers	id	products	seller_id	1:N	CASCADE	Artisan workshop manages multiple craft products in their catalog.
categories	id	products	category_id	1:N	CASCADE	Each product is categorized into a single specific subcategory.
brands	id	products	brand_id	1:N	SET NULL	Artisanal brand tag is optional; deleting brand preserves the product.
orders	id	order_items	order_id	1:N	CASCADE	Master order aggregates itemized order lines across multiple sellers.
products	id	order_items	product_id	1:N	CASCADE	Line item snapshots product ID, title, and agreed selling price.
users (Sender & Receiver)	id	messages	sender_id, receiver_id	1:N	CASCADE	Live chat stream tracking direct communication between customer and artisan.